

Bhavesh Valecha

📍 Augsburg, Germany

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🔗 bhavesh-valecha.github.io

RESEARCH INTERESTS

- Stochastic Thermodynamics
- Nonequilibrium Statistical Physics
- Active Matter
- Stochastic Processes

EDUCATION

Bachelor of Science & Master of Science

📍 Indian Institute of Science Education and Research(IISER) Pune, India

📅 August 2016 - July 2021

CGPA: **8.9/10**

RESEARCH PROJECTS

Thermodynamic Precision in Active Nematics

Supervisor: Prof. Dr. Klaus Kroy, Soft Condensed Matter Theory Group, Universität Leipzig

📅 November 2022 - April 2023

📍 Leipzig, Germany

This project is aimed at, apart from developing other projects, studying the director field precision in the hydrodynamic description of active nematics. In particular, we want to investigate the effect of the striking feature of ‘giant number fluctuations’ in these systems on thermodynamic cost-precision trade-off relations like the TUR.

Violations of the Thermodynamic Uncertainty Relation (TUR)

Supervisors: Dr. Bijay K. Agarwalla, Assistant Professor, IISER Pune & Prof. Dr. Udo Seifert, Director, II. Institute for Theoretical Physics, University of Stuttgart

📅 August 2020 - July 2021

📍 Pune, India

The master’s thesis project focused on the violations of the TUR in open quantum systems and the dependence of such bounds on system size. For a single quantum dot resonant transmission model, we showed that this is indeed the case. We also explored these bounds in the presence of a time-reversal symmetry breaking magnetic field.

Optimizing mesoscopic heat engines(DAAD-WISE Scholarship)

Supervisor: Prof. Dr. Udo Seifert, Director, II. Institute for Theoretical Physics, University of Stuttgart

📅 May 2019 - July 2019

📍 Stuttgart, Germany

The project was aimed at studying the trade-off between the power output, efficiency and the fluctuations in power output using the trajectory approach of Stochastic Thermodynamics and Large Deviation Theory for an engine on the scale of Brownian particles. This trade-off was explored in more detail using the concept of the Pareto front that is commonly used in economics.

Reading on Nonequilibrium Statistical Physics

Supervisor: Prof. Deepak Dhar, Distinguished Visiting Faculty, IISER Pune

📅 Spring 2019

📍 Pune, India

This reading project was aimed at gaining knowledge of the basic ideas and techniques in Nonequilibrium Statistical Physics and to acquire familiarity with associated tools and methods. In particular, the Exclusion Process and the paradoxical Brownian ratchet and pawl systems were studied in detail.

Effects of noise in the initial configuration of growing sandpiles

Supervisor: Prof. Deepak Dhar, Distinguished Visiting Faculty, IISER Pune

📅 May 2017 - December 2018

📍 Pune, India

The primary aim was to study the effects of noise in the initial conditions of the centrally seeded Abelian Sandpile Model(ASM) on a square lattice. We studied patterns generated computationally using numerical methods and found that the average patterns were reminiscent of the well-studied problem of Rayleigh-Benard convection.

CONFERENCE PARTICIPATIONS

WE Heraeus' Quantum Thermodynamics for Young Scientists

📅 02-06 Feb 2020

📍 Bad Honnef, Germany

COURSES TAKEN

PHYSICS

Theory: Classical & Relativistic Mechanics, Waves & Matter, Electrodynamics, Quantum Mechanics I & II, Mathematical Methods in Physics, Statistical Mechanics I & II, Condensed Matter Physics I & II, Quantum Field Theory, General Relativity, Atomic & Molecular Physics.

Extra-curricular: Stochastic Thermodynamics by Prof. Dr. Udo Seifert

Lab: Mechanics & Electromagnetism, Heat & Radiation, Optics, Modern Physics

MATHEMATICS

Single & Multi Variable Calculus, Linear Algebra, Introduction to Proofs, Probability & Statistics, Numerical Analysis, Special Structures in Mathematics, Graph Theory, Algorithms, Data Science

ACADEMIC ACHIEVEMENTS

Scholarships

DAAD-Working Internship in Science and Engineering (WISE) Scholar **2019**

Kishore Vaigyanik Protsahan Yojana (KVPY) Fellow **2017-2021**

DST INSPIRE Scholarship for Higher Education (SHE) **2016-2017**

National Talent Search Examination (NTSE) Scholar **2014-2016**

Talent Camps

VIJYOSHI National Science Camp **2016**

Competitive Exams and Competitions

National Graduate Physics Examination 2017

Kishore Vaigyanik Protsahan Yojana(KVPY) 2016

Joint Entrance Exam (Mains) 2016

Maharashtra Common Entrance Test

State Topper

All India Rank 35

All India Rank 1773

Scored 189/200

PROGRAMMING

C, Python, FORTRAN

EXTRA-CURRICULAR ACTIVITIES

Sports

Part of Football Team - Reached the semi-final of Intra IISER Football League 2018

Music

Can play the Ukulele

Community Service

Part of the 'Spread The Smile' Initiative 2017 for teaching kids in remote villages

Miscellaneous

- IISER Pune Quiz Club Coordinator (2017-18)

- Part of the Core Team of Physics question making body for a national level quiz conducted by IISER Pune called 'Mimamsa'

REFERENCES

- Prof. Dr. Abhinav Sharma
- Prof. Dr. Udo Seifert

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